

The design of this cubic chip (not to be confused with the Power Mac Apple G4 cube, which was a computer in itself) is purportedly the first to integrate each layer in such an optimally seamless and efficient manner. Piling several integrated circuits together made it necessary to first ensure an effective insulation between each chip and then drill thousands of perforations in the insular layer to allow vertical connectivity. The prototype of this 'cube' is already functioning at the University of Rochester at a speed of 1.4 gigahertz.

Eby Friedman, distinguished Professor of Electrical and Computer Engineering at Rochester University, New York, USA, is the Director of this project and he's had help in the form of engineering student Vasilis Pavlidis. The chip, which has been specially fabricated at MIT (Massachusetts Institute of Technology), is still in the prototype stage.

And If It Comes Through...

The continuous shrinking of integrated circuits augments speed but connecting multiple chips horizontally means that more space is required. Since all the layers act like a single system, the Rochester chip functions like a folded-up circuit board. Imagine the motherboard of you computer shrunk to the size of a Rubik's cube. Besides, the architecture of the cube is such that it could increase the speed of your iPod or cell phone by up to ten times that of chips today. More height means less width, so finally perhaps, we'll have flatter CPUs, smaller printers, miniscule iPods etc. — and as a result, more space to use around the room. Skepticism has been voiced on whether the industry would take to it well, but we at *Digit* feel that the future belongs to chipper chips and not whopper circuit boards.

2. PROTEIN SHAKERS

Hard Disks Go Organic

Imitating Nature

If nature can use proteins to help our brains store memory, why can't we? If you take a good look at our CDs and DVDs, you'll realise that they are enhancements of the vinyl record – albeit on a microscopic scale. Perhaps then, our use of synthetic materials is due to end. Memory-based devices made out of biological materials have long since been considered to have the power to process information more quickly and allow more data to be stored than the present-day options available to us. Several experiments in the past

have floundered but the challenge draws humankind onwards.

Amiable Aminos

Our traditional hard disks, CDs, etc. are either magnetic or optical data storage systems which are becoming, well, harder disks to put more data into. Rooms full of memory devices seem to be the only way of managing the mammoth databases the world is now dealing with. The most significant advancement in this area has been made by researchers in Japan who have managed to develop a new protein-based memory device.

Koji Nakayama, Tetsuro Majima and Takashi Tachikawa have succeeded in etching or 'recording' specific data on a glass slide, using a fluorescent protein. The combined use of light and chemicals effectively stored information that was later 'read' and then erased. Thus, recording, playback and deletion – the basic functions of memory storage instruments – were proved possible using biological materials. They define the material as 'a biological device that enables us to spatiotemporally photoregulate the recording, reading, and erasing of information on a solid surface using protein'.

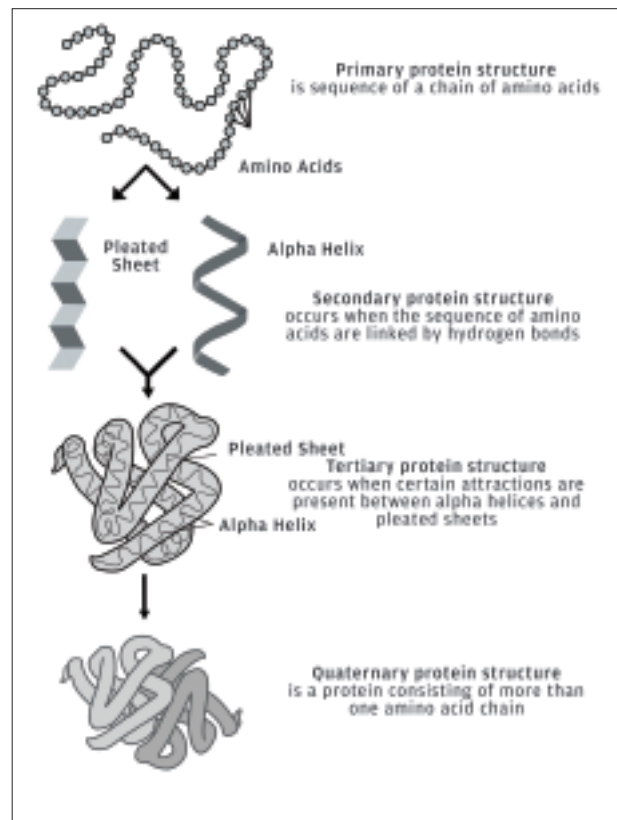
And If It Comes Through...

The scientists involved in the project have themselves suggested that the technology could be used for biosensing and diagnostic

A chip off the old block



Dr Eby Friedman



The Structure of Proteins

assays. But of importance to us is their third suggestion, that it be utilised for 'record-erasable soft material'. The possibilities, if this works, are limitless, and a reversal of sorts, where the bio-chip seamlessly entering our bodies to enhance human functionality seems possible in the not-so-distant future. Contemplating the consequences of bio-memory instruments, one immediately fears 'Terminator 3' and 'Matrix'-like dystopian scenarios, but never fear, the research is not even in the prototype stage and has a long way to go before it is proved to work as well as today's storage instruments. Several other simultaneous efforts towards the protein chip are ongoing and therefore it remains to be seen who will come up with the best product.

3. SENSOR GLOVES

Reckoned Skin

More Than Just Haptic

From the calculator watch to the HMDs (Helmet Mounted Displays), we have always been a little impatient and have now firmly begun to believe that a person's computer should be worn, much like eyeglasses or clothing are worn, and interact with the user based on the context of the situation. In fact, at a time when skin is being treated more and more like cloth, intelligent clothes are one sure-shot way to bring back to clothes their primal status of functional accessories. While laptops and palmtops are steps in this direction, serious advances indicate that the dream may not remain a dream much longer.

Fits Like A Glove

Most technological advancements and breakthroughs, regrettably, emerge from conflict, war and the needs of the military (ARPANET, aviation technology, etc.). The latest example is an intelligent glove. US soldiers in Iraq already use wearable computer systems but lack efficient input devices. Now, a company called RallyPoint, based in Cambridge, MA, has developed a sensor-embedded glove that allows the soldier to easily view and navigate using digital maps, activate radio communications, and send commands without needing to use his hands. This isn't so great when you consider that several groups have been working and bringing out sensor filled gloves in the past, using accelerometers, gyroscopes, and other high-tech sensors.

However, this one is a little different, because it is more practical, rugged and made for the military. It has been designed in such a way that a soldier can use it to grip an object and still continue to use its electronic capacities. The glove has four custom-built push-button sensors



Hand in glove

sewn into the fingers. Radio can be activated by the sensors on the tips of the middle and fourth fingers, each finger used to locate a different channel. On the lower portion of the index finger is a tiny sensor that can help change modes, from "map mode" to "mouse mode". Another sensor, on the little finger, can be used to zoom in (or out) of a map, while in 'Map Mode'. The same sensor in 'Mouse Mode' is a mouse-click button.

And If It Comes Through...

Although it probably really hasn't been envisaged yet, the glove-computer has immense possibilities for the future of gaming. We all know about the magic of the Apple Motion Sensor, PSP Sixaxis motion detection etc. But with the Glove-computer, the extent of immersion and interaction into the game could increase ten-fold. No more handheld pads or joystick surrogates. Everything you need would literally be in you hands. Now if only they could find a way to make it wireless...

4. ANTI-VIRUS CLUSTER

A Cloud Full Of Silver Linings

Cloud Computing

If you've heard of Web 2.0, no doubt you've heard of Cloud Computing – but we'll tell you anyway. Cloud computing is basically a concept that involves Web 2.0, SaaS and the latest trends in technology to provide seamless, better enriched services using the internet. No self-respecting computer today can get by without a good anti-virus software installed and trying to grapple with the number of trojans, malware, worms and hacker-go-lucky viruses that are trying to infiltrate into your sytem.

Remote Control

How nice it would be if the task of checking the files and documents that you open was done by some software deep in the infinite web, which monitors your PC remotely! Researchers at the University of Michigan developed a new cloud-based approach to antivirus which they call 'CloudAV' and which, they claim, can outdo any anti-virus package on the market.

Prof. Farnam Jahanian, professor of computer science and engineering in the Department of Electrical Engineering and